

● EASY

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

08

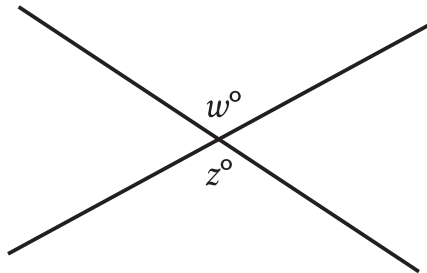
7 questions · ~11 min

SAT QUESTION BANK

2026-05-29

Q1

GEOMETRY AND TRIGONOMETRY

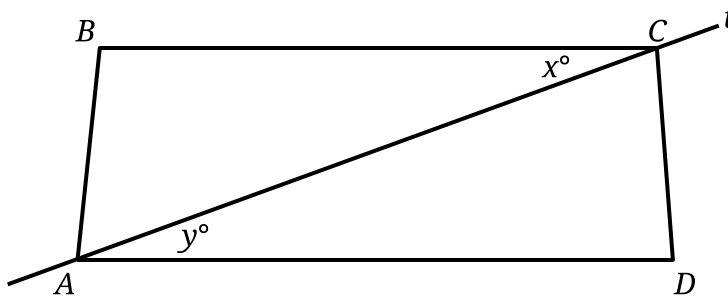


Note: Figure not drawn to scale.

- At the intersection of the 2 lines, the angles are labeled clockwise from top as follows:
 - Top: w°
 - Right: unlabeled
 - Bottom: z°
 - Left: unlabeled
- A note indicates the figure is not drawn to scale.

In the figure, two lines intersect at a point. If $w = 136$, what is the value of z ?

- A. 36
- B. 44
- C. 68
- D. 136

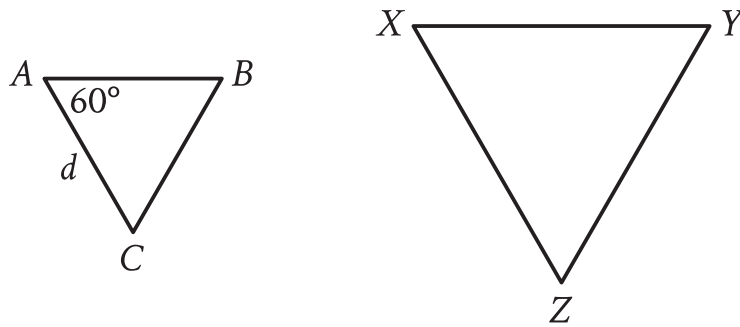


Note: Figure not drawn to scale.

- Line t passes through points upper A and upper C.
- The measure of angle upper A upper C upper B is x° .
- The measure of angle upper C upper A upper D is y° .
- A note indicates the figure is not drawn to scale.

In the figure shown, $\overline{BC}$ is parallel to $\overline{AD}$ and $x = 22$. Line t passes through points A and C . What is the value of y ?

- A. 22
- B. 44
- C. 68
- D. 158



Note: Figures not drawn to scale.

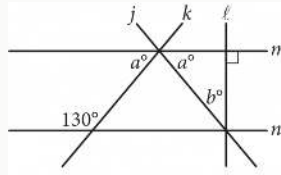
- The 2 triangles are oriented such that upper A corresponds to upper X, upper B corresponds to upper Y, and upper C corresponds to upper Z.
- Triangle upper A upper B upper C is labeled as follows:
 - The measure of angle upper A is 60° .
 - The length of side upper A upper C is d .
- A note indicates the figures are not drawn to scale.

For the triangles shown, triangle ABC is dilated by a scale factor of 3 to obtain triangle XYZ , where $d = 16$. What is the measure, in degrees, of angle X ?

- A. 20
- B. 57
- C. 60
- D. 63

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 18° . What is the measure of angle F ?

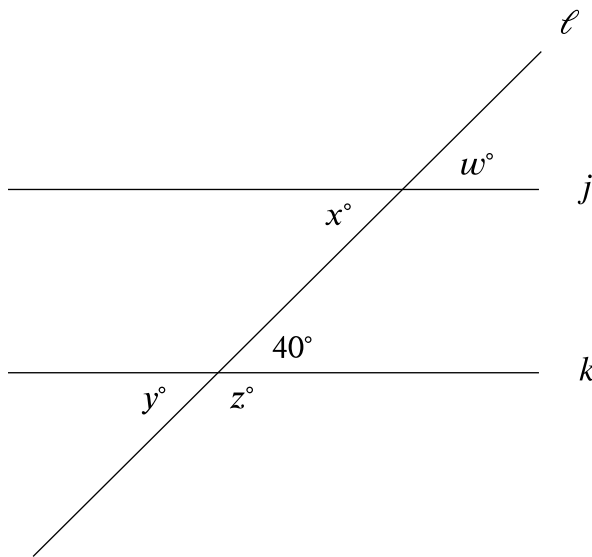
- A. 18°
- B. 72°
- C. 90°
- D. 162°



Note: Figure not drawn to scale.

In the figure above, lines m and n are parallel. What is the value of b ?

- A. 40
- B. 50
- C. 65
- D. 80



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled script l , j , and k .
- Line script l intersects both line j and line k .
- At the intersection of line script l and line j , 2 angles are labeled clockwise from top left as follows:
 - Top right: w°
 - Bottom left: x°
- At the intersection of line script l and line k , 3 angles are labeled clockwise from top left as follows:
 - Top right: 40°
 - Bottom right: z°
 - Bottom left: y°
- A note indicates the figure is not drawn to scale.

In the figure shown, line l intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

A. $w = 40$

B. $x = 140$

C. $y = 40$

D. $z = 140$

In $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98° . What is the measure of $\angle Z$?

- A. 58°
- B. 74°
- C. 122°
- D. 212°